IE221 - Kỹ thuật lập trình Python

BUỔI 4: Bài thực hành 5

TH X O NUMPY

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NumPy là gì?

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- N o o o ck s o s o S ck
  o n

D ch ch O

```
import numpy as os
```

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NumPy Array vs Python List

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```
In [3]:  a = [1, 2, 3]
         [q*2 for q in a]
Out[3]: [2, 4, 6]
```

```
In [4]:  a = np.array([1, 2, 3])
         a * 2
Out[4]: array([2, 4, 6])
```

```
In [1]:  a = [1, 2, 3]
         b = [4, 5, 6]
         [q+r for q, r in zip(a, b)]
Out[1]: [5, 7, 9]
```

```
In [2]:  a = np.array([1, 2, 3])
         b = np.array([4, 5, 6])
         a + b
Out[2]: array([5, 7, 9])
```

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NumPy Array vs Python List

ch ch NumPy

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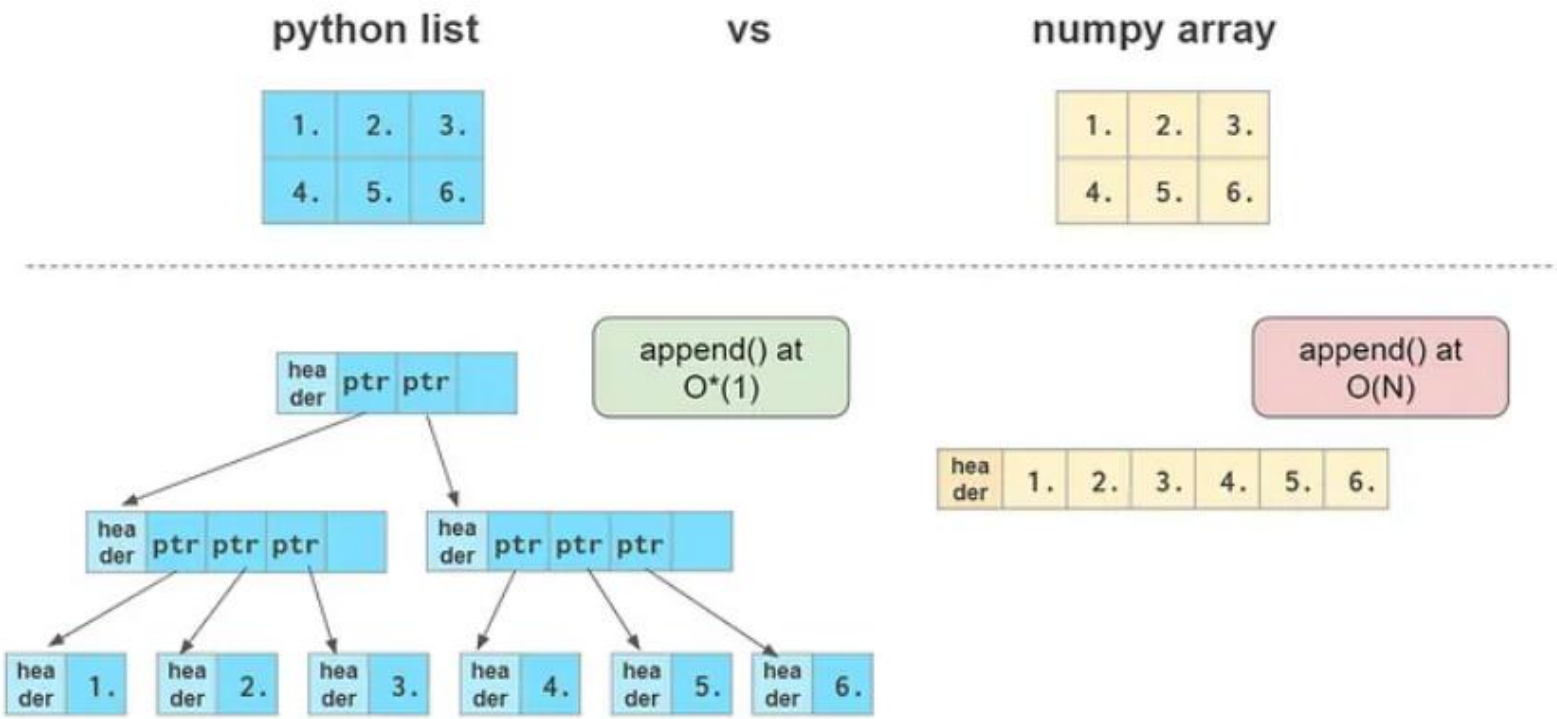

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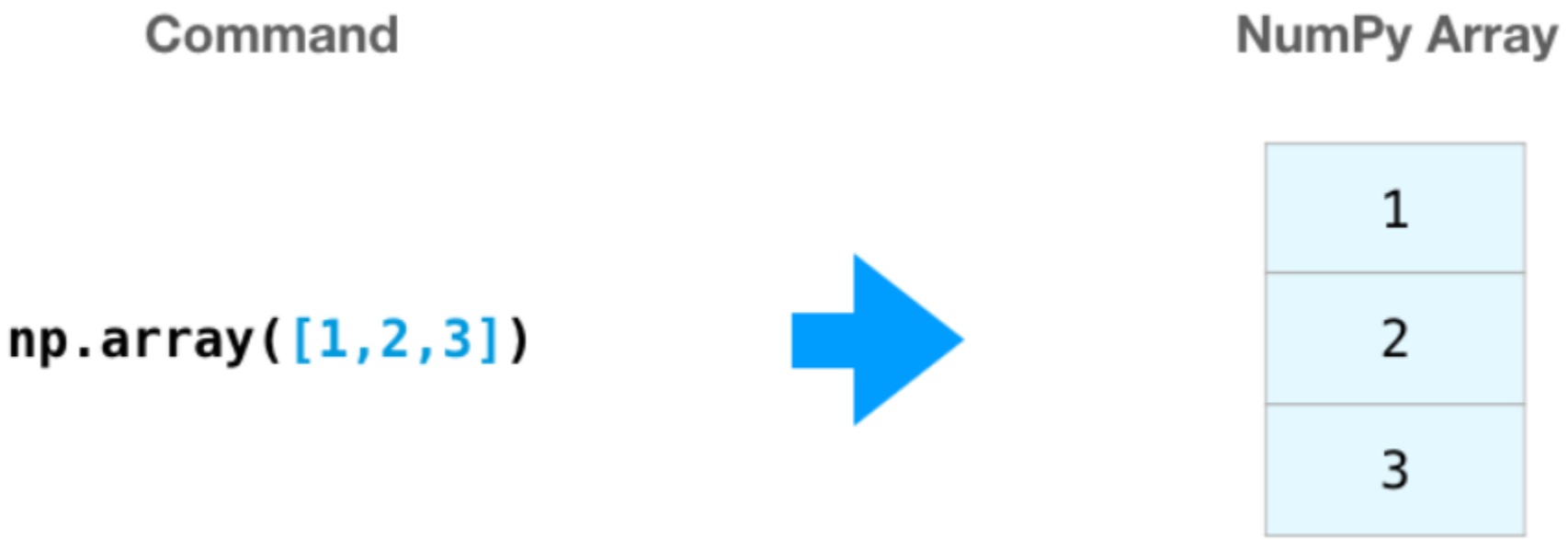
D chtich

Chúng ta có thể tạo một mảng NumPy (hay còn gọi là **ndarray**) bằng cách truyền **một danh sách** Python vào nó và sử dụng `**np.array()**` . Trong trường hợp này, Python tạo ra mảng mà chúng ta có thể thấy ở bên phải.



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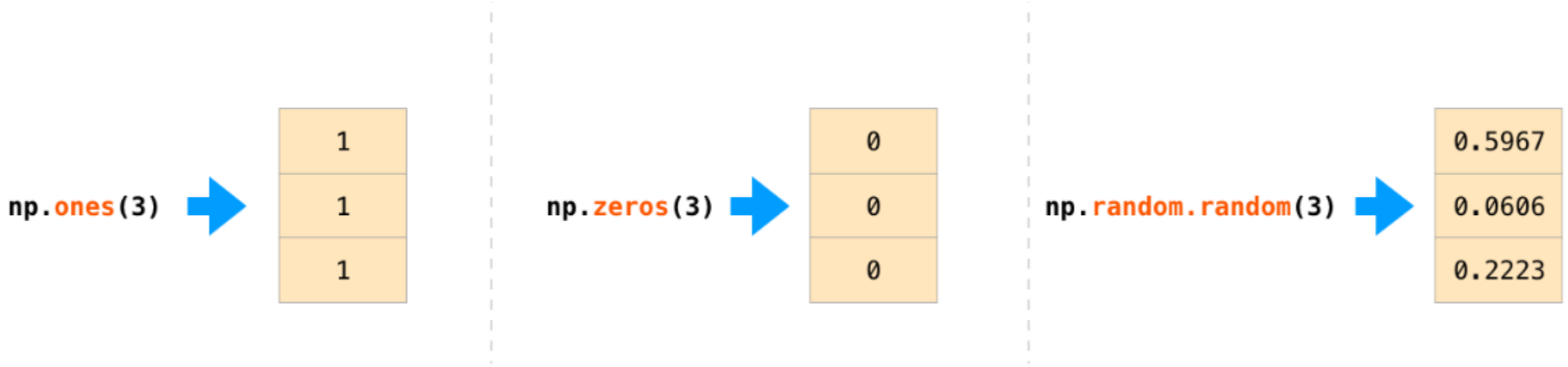
D chtch

Thường có những trường hợp chúng ta muốn NumPy khởi tạo các giá trị của mảng cho mình. NumPy cung cấp các phương thức như **ones(), zeros()** và **random.random()** cho những trường hợp này. Chúng ta chỉ cần truyền cho chúng số lượng phần tử mà chúng ta muốn nó tạo ra



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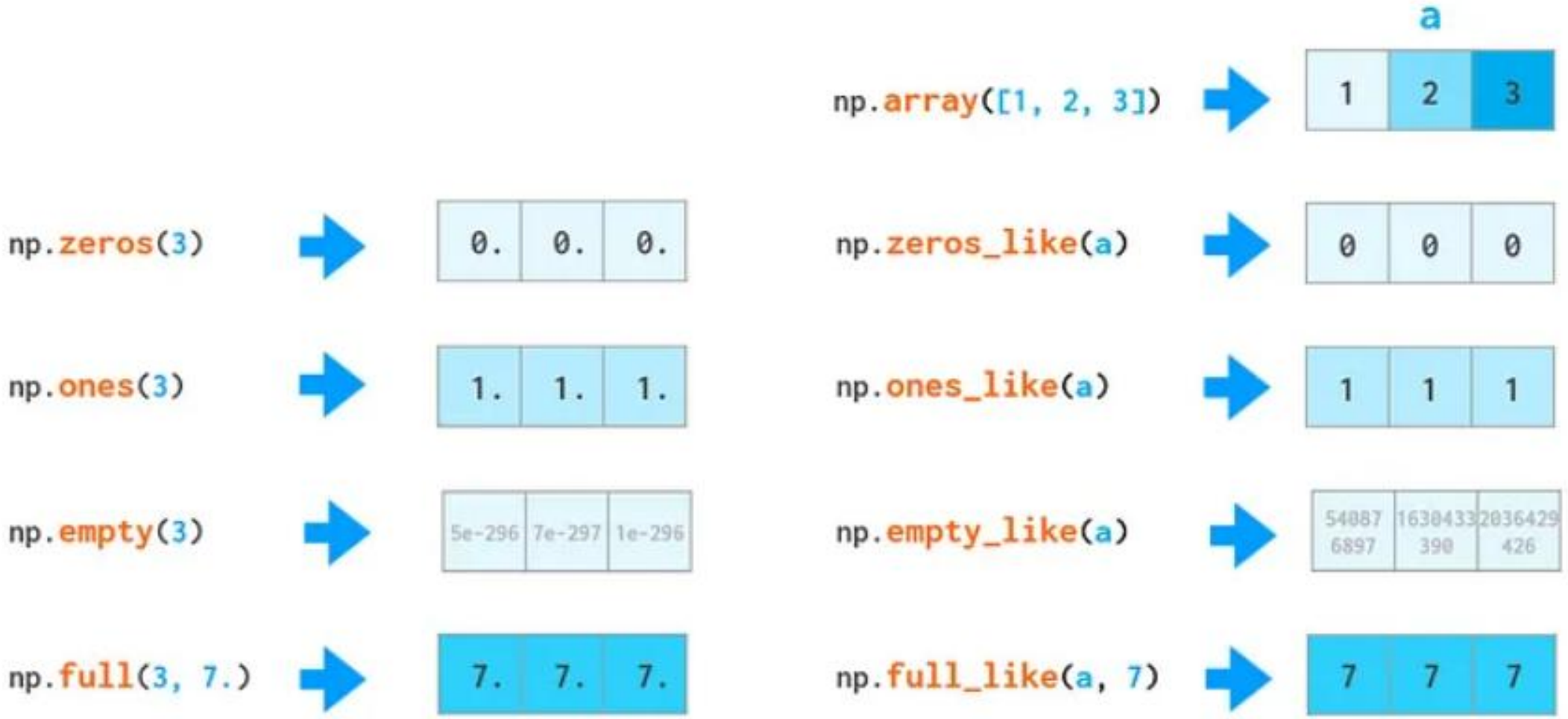
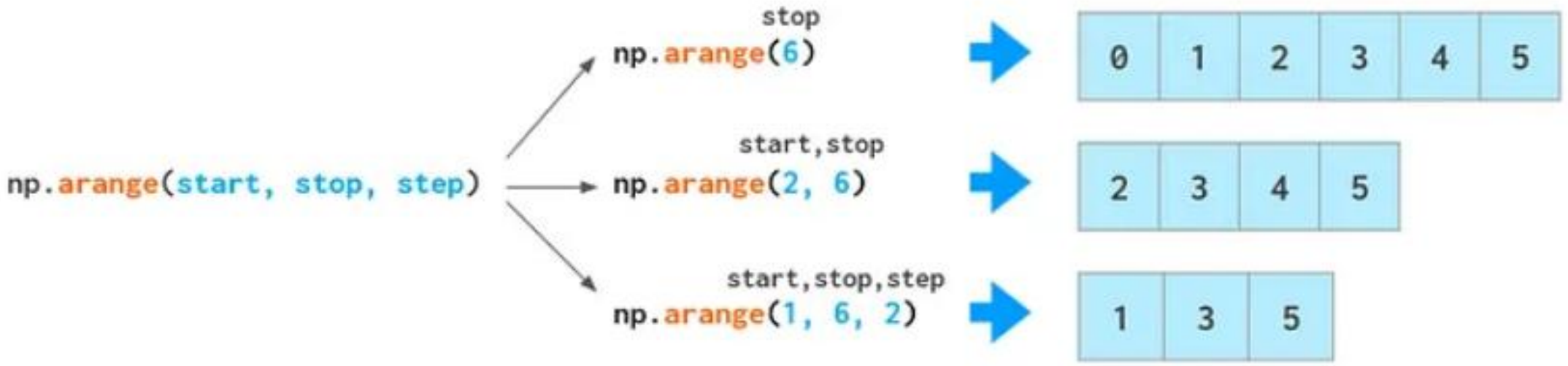
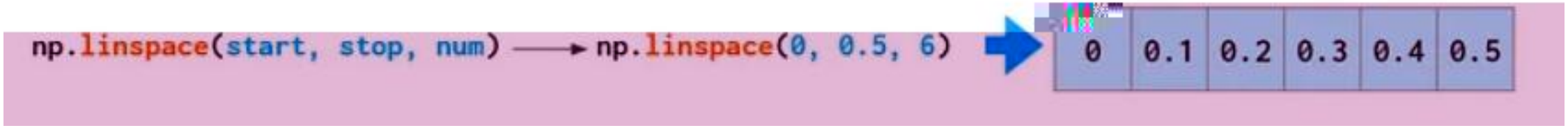

a
np.array([1, 2, 3])
1 2 3
np.zeros(3)
0. 0. 0.
np.zeros_like(a)
0 0 0
np.ones(3)
1. 1. 1.
np.ones_like(a)
1 1 1
np.empty(3)
5e-296 7e-297 1e-296
np.empty_like(a)
54887 6897 1630433 390 2036429 426
np.full(3, 7.)
7. 7. 7.
np.full_like(a, 7)
7 7 7

O

D chchch

Có tới hai hàm để khởi tạo mảng với chuỗi đơn điệu trong NumPy:





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Nhưng arange không thực sự tốt trong việc xử lý số float:

- norm: `np.arange(0.4, 0.8, 0.1)` (start stop step) → 0.4 | 0.5 | 0.6 | 0.7
- anomaly: `np.arange(0.5, 0.8, 0.1)` → 0.5 | 0.6 | 0.7 | 0.8
- solution1: `np.arange(0.5, 0.75, 0.1)` → 0.5 | 0.6 | 0.7
- solution2: `np.linspace(0.5, 0.7, 3)` (start stop num) → 0.5 | 0.6 | 0.7
- gotcha: `np.linspace(0, 1, ?)` → 0. | 0.1 | 0.2 | 0.3 | ... | 1.

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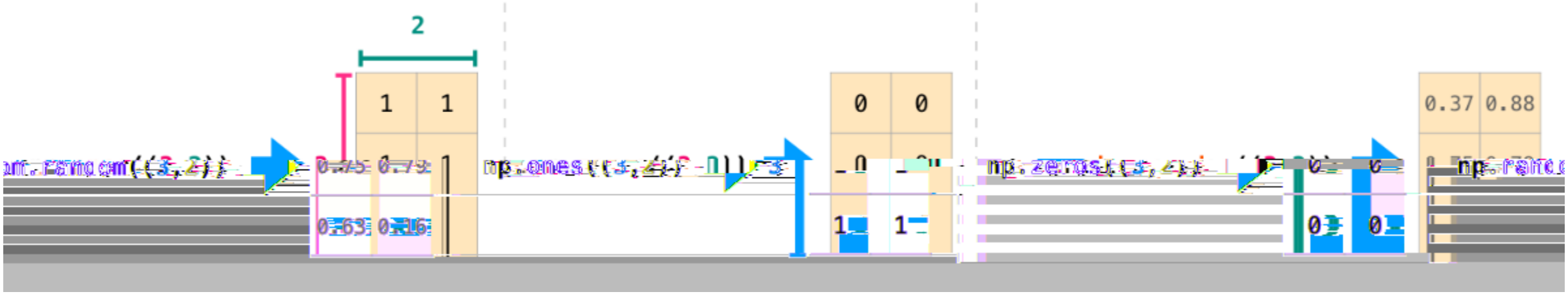
```
np.array([[1,2],[3,4]])
```

1	2
3	4



o

Shape và Dimensions

shape: sn ck ıck ưck ck

ndim: ck ck o

size: o s o

dtype: n ck s o

```
a = np.array([1., 2., 3.])
```

```
b = np.zeros(3, int)
```

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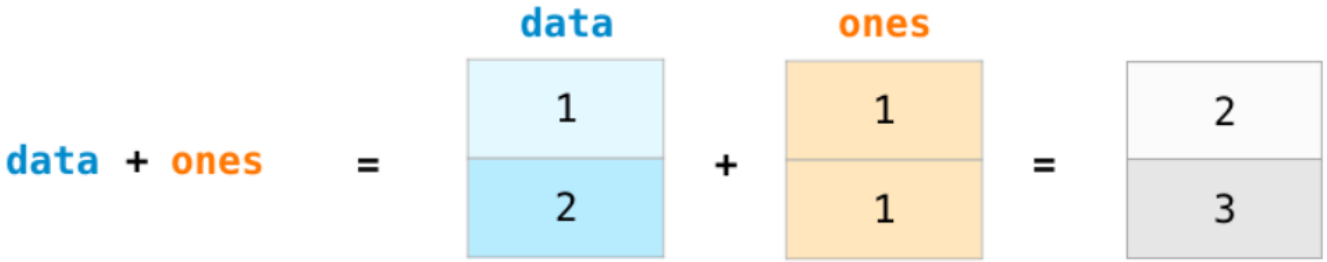
Element-wise operations (từng phần tử):

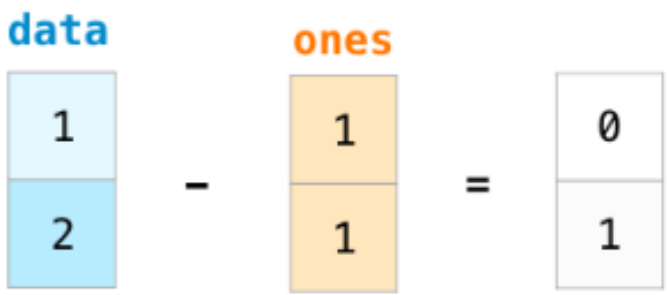
D `a + b` `np.add(a, b)`

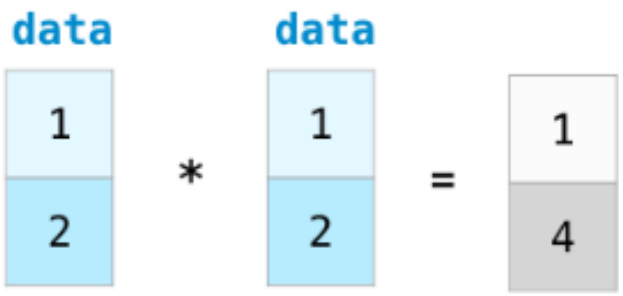
`a - b` `np.subtract(a, b)`

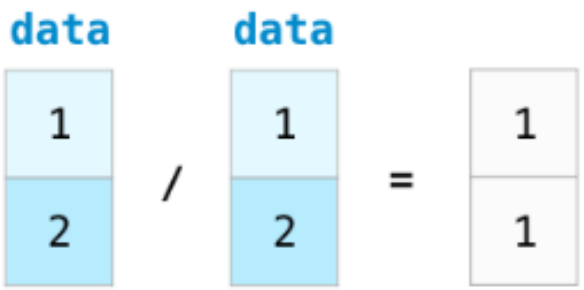
Nhân `a * b` `np.multiply(a, b)`

Chia `a / b` `np.divide(a, b)`









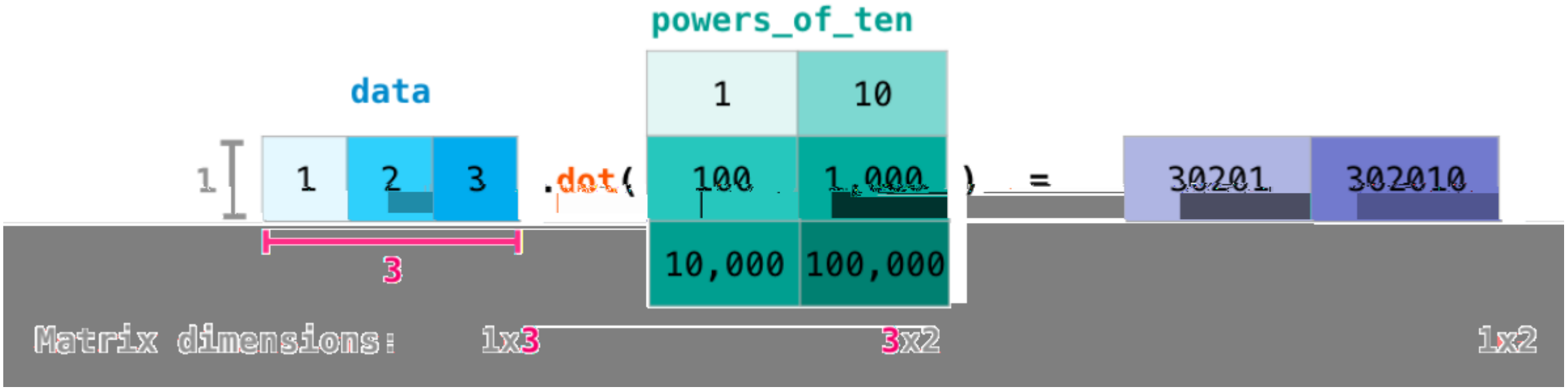
O D

ch np.dot(a, b)

O a @ b ch np.matmul(a, b)

D a.T ch np.transpose(a)

Outer product np.outer(a, b)



O D

Universal Functions (ufuncs)

ch

`np.sqrt()` `np.exp()` `np.log()`

M ch `np.sin()` `np.cos()`

So sánh `np.maximum()`

O 4: INDEXING & SLICING

Indexing cơ bản

o ch

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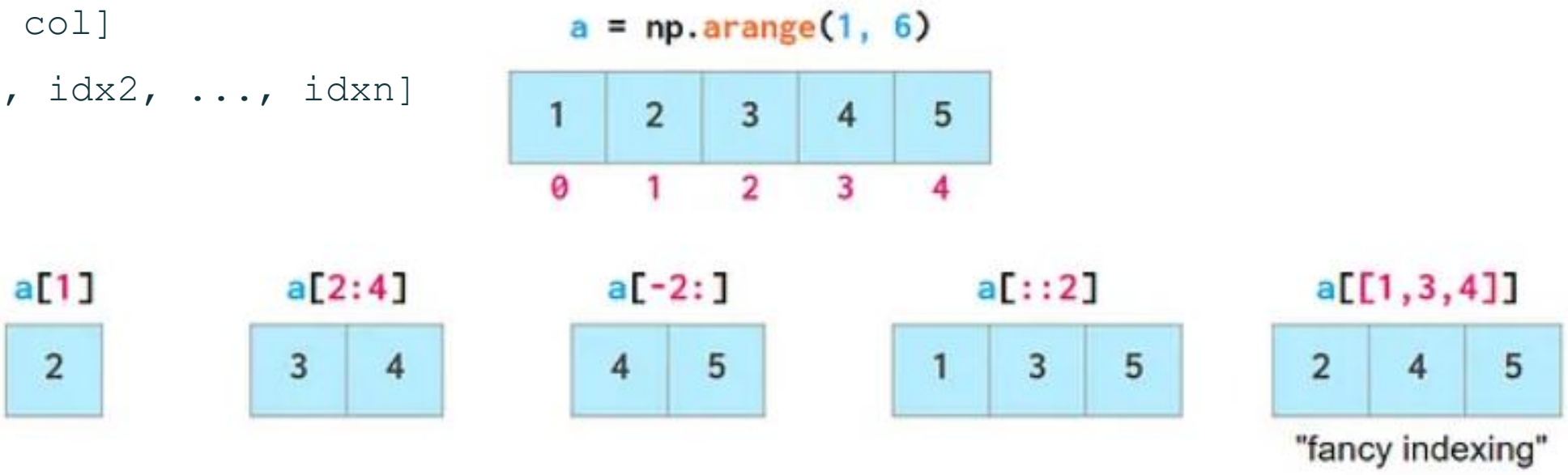
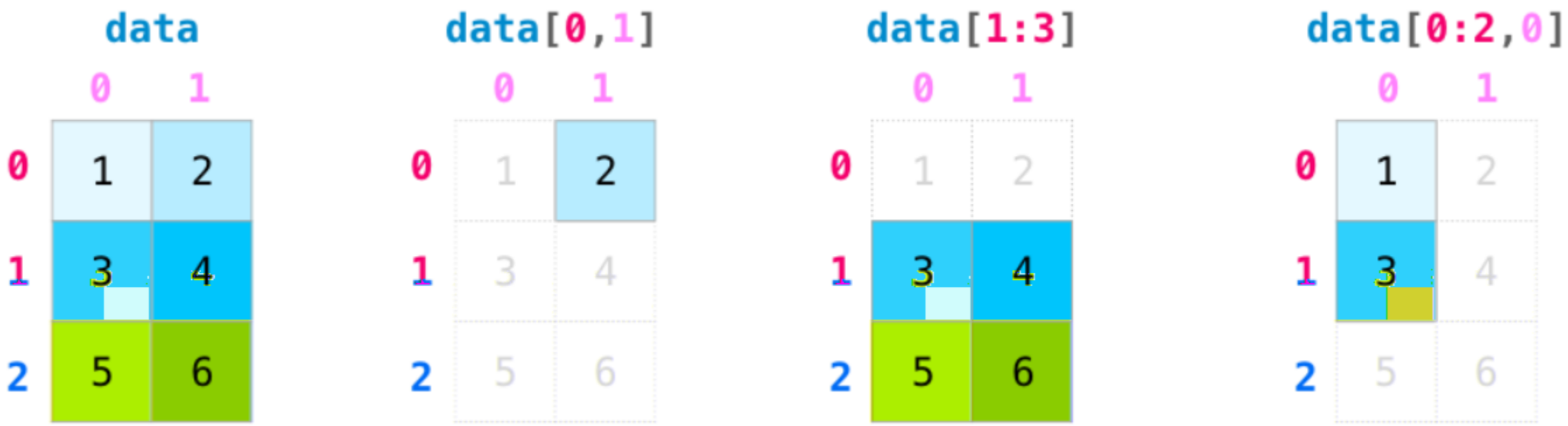
```
arr[index]
arr[row, col]
```

o `arr[idx1, idx2, ..., idxn]`



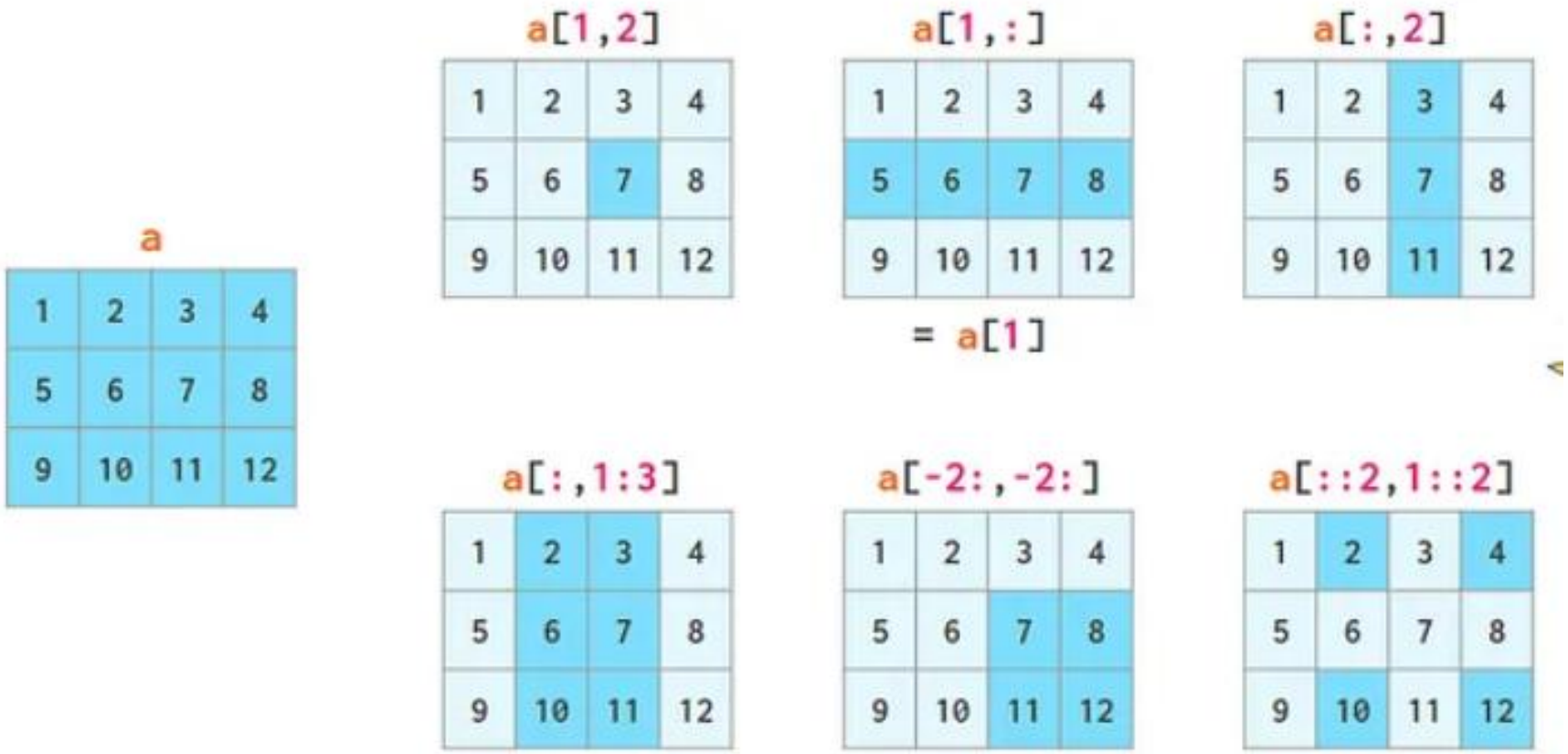
O 4: INDEXING & SLICING

Indexing cơ bản



04: INDEXING & SLICING

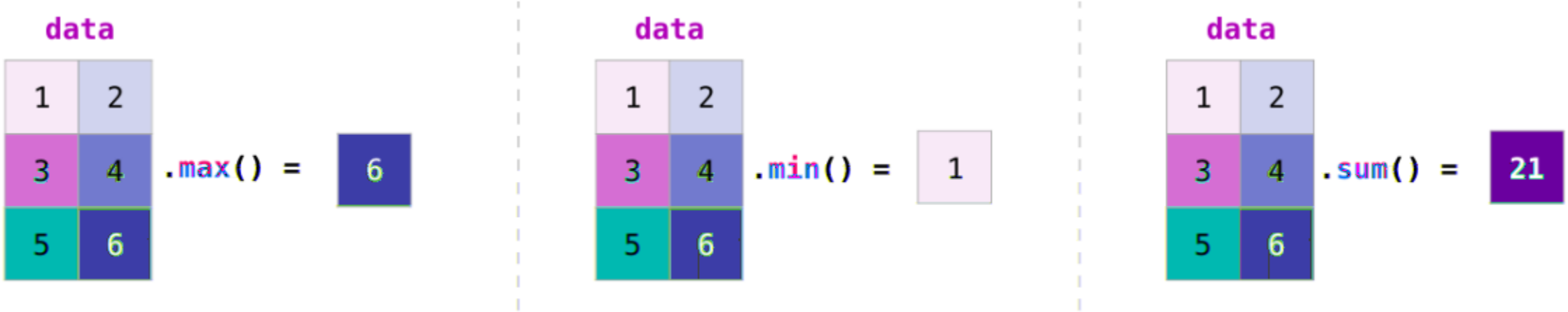
Slicing và Advanced Indexing



O 5: AGGREGATION

Aggregation Functions

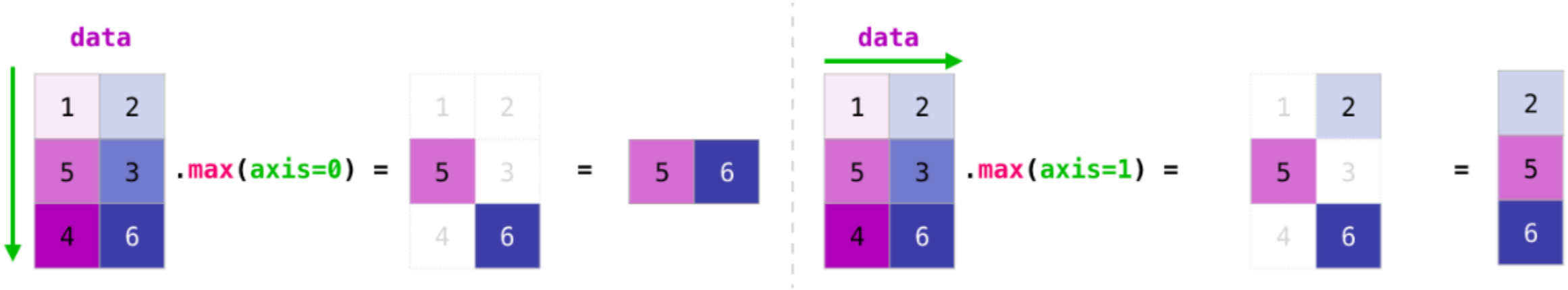
```
np.sum(arr)    o ckcks   o
np.mean(arr)    o cho
np.min() np.max()       o   nưo o
np.std() np.var()    n ck ck   o s    o
      axis   o )   o ch   )    ck    )       o
```



O 5: AGGREGATION

Aggregation Functions

```
np.sum(arr)    o ckcks   o
np.mean(arr)    o cho
np.min() np.max()       o   nưo o
np.std() np.var()    n ck ck   o s    o
      axis   o )   o ch   )    ck    )       o
```



05: AGGREGATION

Aggregation Functions

O 6: TRANSPOSE VÀ RESHAPE

Transpose

o o y ck ck o

arr.T ck np.transpose(arr)

s) o)o

o ơo o n oy ON

data

1	2
3	4
5	6

data.T

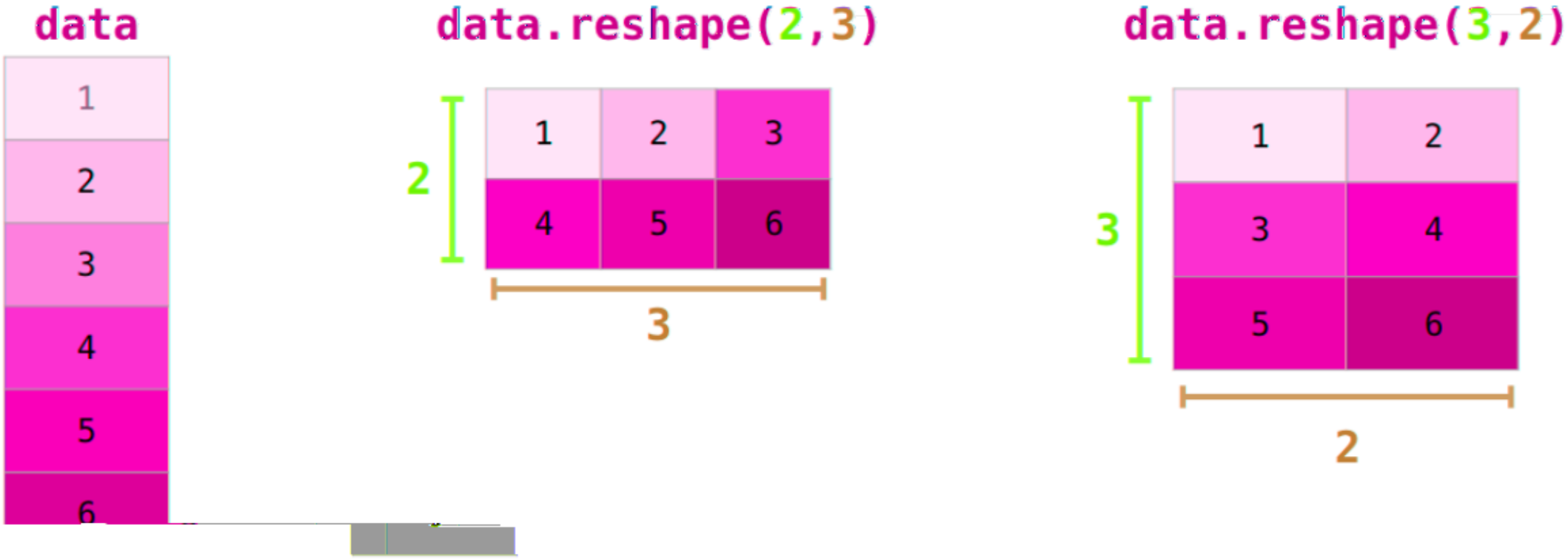
1	3	5
2	4	6

O 6: TRANSPOSE VÀ RESHAPE

Reshape

s o n

```
arr.reshape(new_shape)
```



O O O D - FORMULAS

Vector hóa công thức - MSE

Mean Squared Error (MSE):

o ck

$$MSE = \frac{1}{n} * \sum (y_pred - y_true)^2$$

```
error = (1/3) * np.sum(np.square(predictions - labels))
```

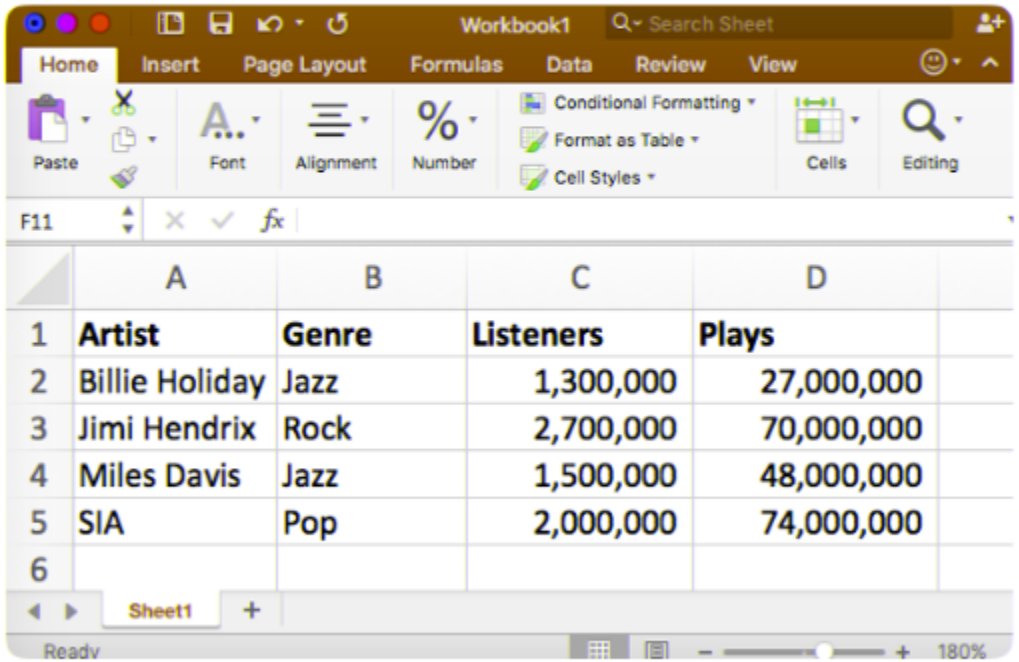
predictions labels

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

O O O D - TABLES

Bảng dữ liệu - 2D Arrays

music.csv



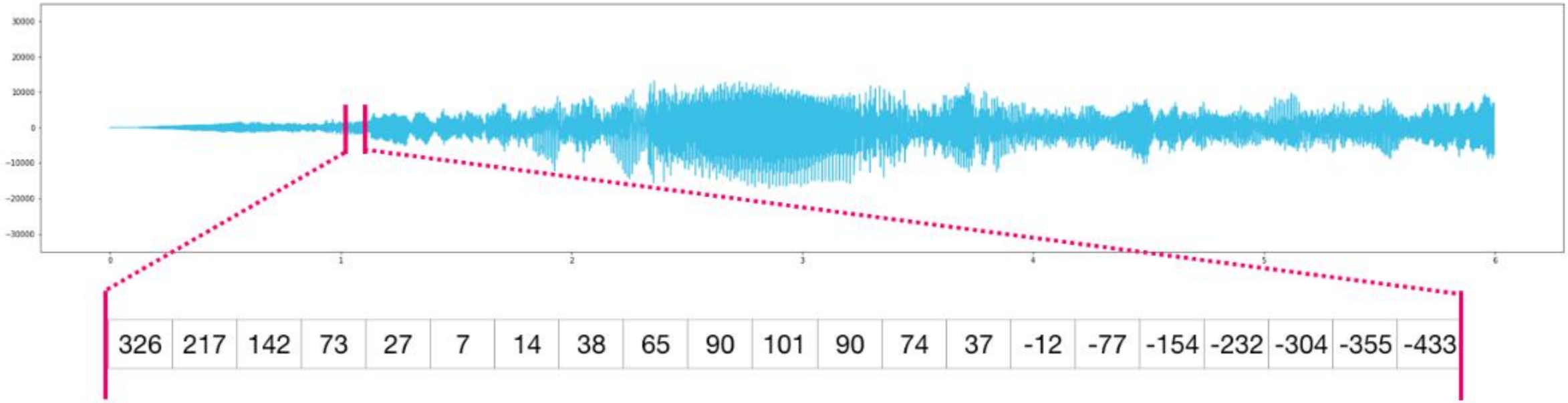
	A	B	C	D
1	Artist	Genre	Listeners	Plays
2	Billie Holiday	Jazz	1,300,000	27,000,000
3	Jimi Hendrix	Rock	2,700,000	70,000,000
4	Miles Davis	Jazz	1,500,000	48,000,000
5	SIA	Pop	2,000,000	74,000,000
6				

pandas.read_csv('music.csv')

	Artist	Genre	Listeners	Plays
0	Billie Holiday	Jazz	1,300,000	27,000,00
1	Jimi Hendrix	Rock	2,700,000	70,000,00
2	Miles Davis	Jazz	1,500,000	48,000,00
3	SIA	Pop	2,000,000	74,000,00

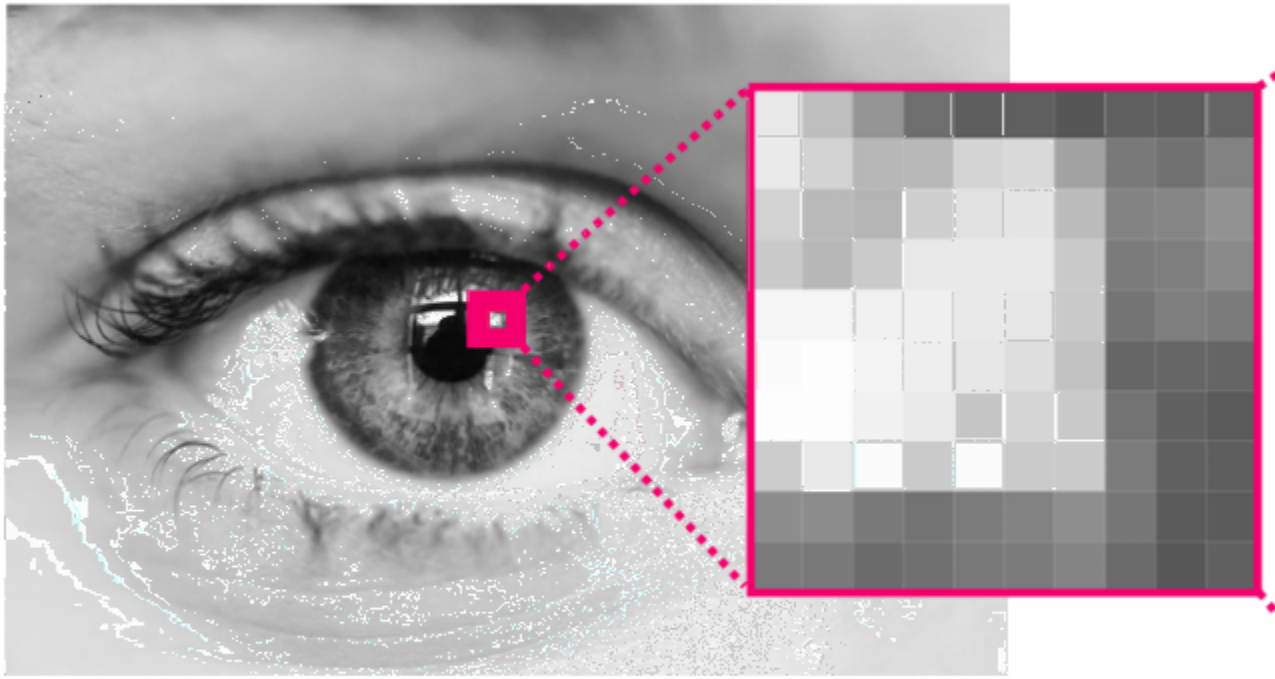
O 8: DATA REPRESENTATION - AUDIO

Audio - Mảng 1D và 2D



O 8: DATA REPRESENTATION - IMAGES

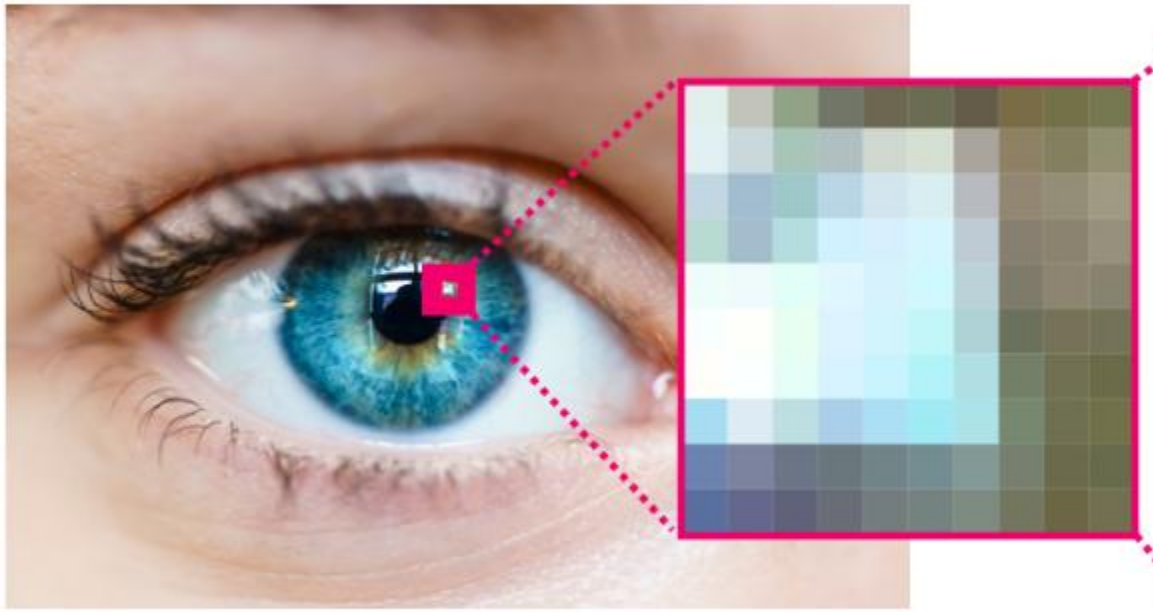
Hình ảnh - Mảng 2D và 3D



230	194	147	108	90	98	84	96	91	101
237	206	188	195	207	213	163	123	116	128
210	183	180	205	224	234	188	122	134	147
198	189	201	227	229	232	200	125	127	135
249	241	237	244	232	226	202	116	125	126
251	254	241	239	230	217	196	102	103	99
243	255	240	231	227	214	203	116	95	91
204	231	208	200	207	201	200	121	95	95
144	140	120	115	125	127	143	118	92	91
121	121	108	109	122	121	134	106	86	97

O 8: DATA REPRESENTATION - IMAGES

Hình ảnh - Mảng 2D và 3D



		233	188	137	96	90	95	63	73	73	82
	237	202	159	120	105	110	88	107	112	121	109
226	191	147	110	101	112	98	123	110	119	142	131
221	191	176	182	203	214	169	144	133	145	155	122
185	160	161	184	205	223	186	137	147	161	140	115
181	174	189	207	206	215	194	136	142	151	133	87
246	237	237	231	208	206	192	122	143	144	111	74
254	254	241	224	199	192	181	99	122	117	107	74
239	248	232	207	187	182	184	110	114	110	113	74
193	215	193	167	158	164	181	114	112	111	105	82
113	119	110	111	113	123	135	120	108	106	113	
93	97	91	103	107	111	122	112	104	114		

O 8: DATA REPRESENTATION - TEXT

Văn bản - Embeddings & Vectors

b

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yco O o s)o ch o

Model Vocabulary

#	
0	the
1	of
2	and
…	…
71,289	dolophine

have	the	bards	who	preceded	me	left	any	theme	unsung
38	0	29104	56	7027	745	225	104	2211	66609

O 8: DATA REPRESENTATION - TEXT

Batch Processing - NLP

D ck ckckck

s)ch ck ª n o ch o

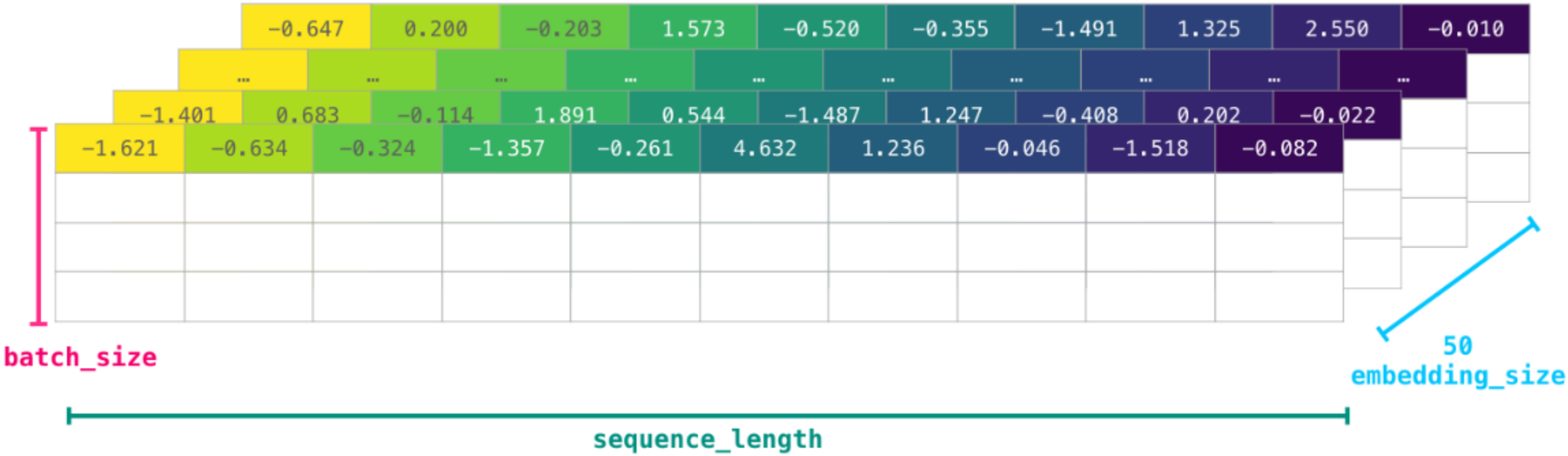
S o o o

	have	the	bards	who	preceded	me	left	any	theme	unsung
0	-1.621	-0.634	-0.324	-1.357	-0.261	4.632	1.236	-0.046	-1.518	-0.082
1	-1.401	0.683	-0.114	1.891	0.544	-1.487	1.247	-0.408	0.202	-0.022
	…	…	…	…	…	…	…	…	…	…
49	-0.647	0.200	-0.203	1.573	-0.520	-0.355	-1.491	1.325	2.550	-0.010

O 8: DATA REPRESENTATION - TEXT

Batch Processing - NLP

D ck ckckck

s)ch ck ª n o ch o

S o o o